CHAPTER 2
THERMOCHEMISTRY

2.1 CONCEPT OF ENTHALPY

THERMOCHEMISTRY

- All chemical reactions involve bond breaking in the reactant molecules (energy is absorbed) and bond formation in the product molecules (energy is released).
- These chemical reactions involve changes in energy.
- The study of heat change in chemical reactions is called **thermochemistry**.

Enthalpy (H)

- **Enthalpy** is the heat content of a system at constant pressure.
- The enthalpy of a system **cannot be measured directly** but the enthalpy change (ΔH) that takes place in chemical reaction can be measured.

Enthalpy Change of Reaction (ΔH)

- **Enthalpy change** is the amount of heat absorbed or released by a system in a chemical reaction at constant pressure.
- It can be expressed as follows :

$$\Delta H_{rxn} = \sum mH^o_f \text{ products} - \sum nH^o_f \text{ reactants}$$

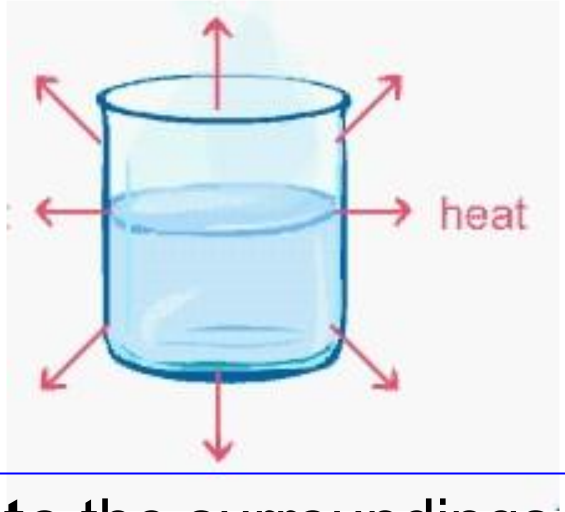
- **Standard enthalpy change (ΔH°)** is amount of heat absorbed @ released by a system in chemical reaction under standard condition.

- Standard conditions : T = 298.15K, P = 1 atm, concentration of an aqueous solution = 1 M

* (all substances presence under the most stable physical state at 25^oC and 1 atm).

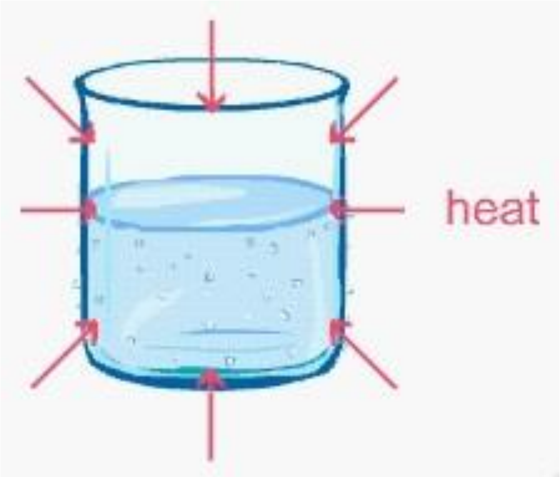
Two types of enthalpy change in chemical reactions

Exothermic reaction



- Heat is **released to** the surroundings.
- Temperature of the surrounding **increases**.
- $\sum H_{products}$ **<** $\sum H_{reactants}$
- ΔH **< 0** (**negative value**)

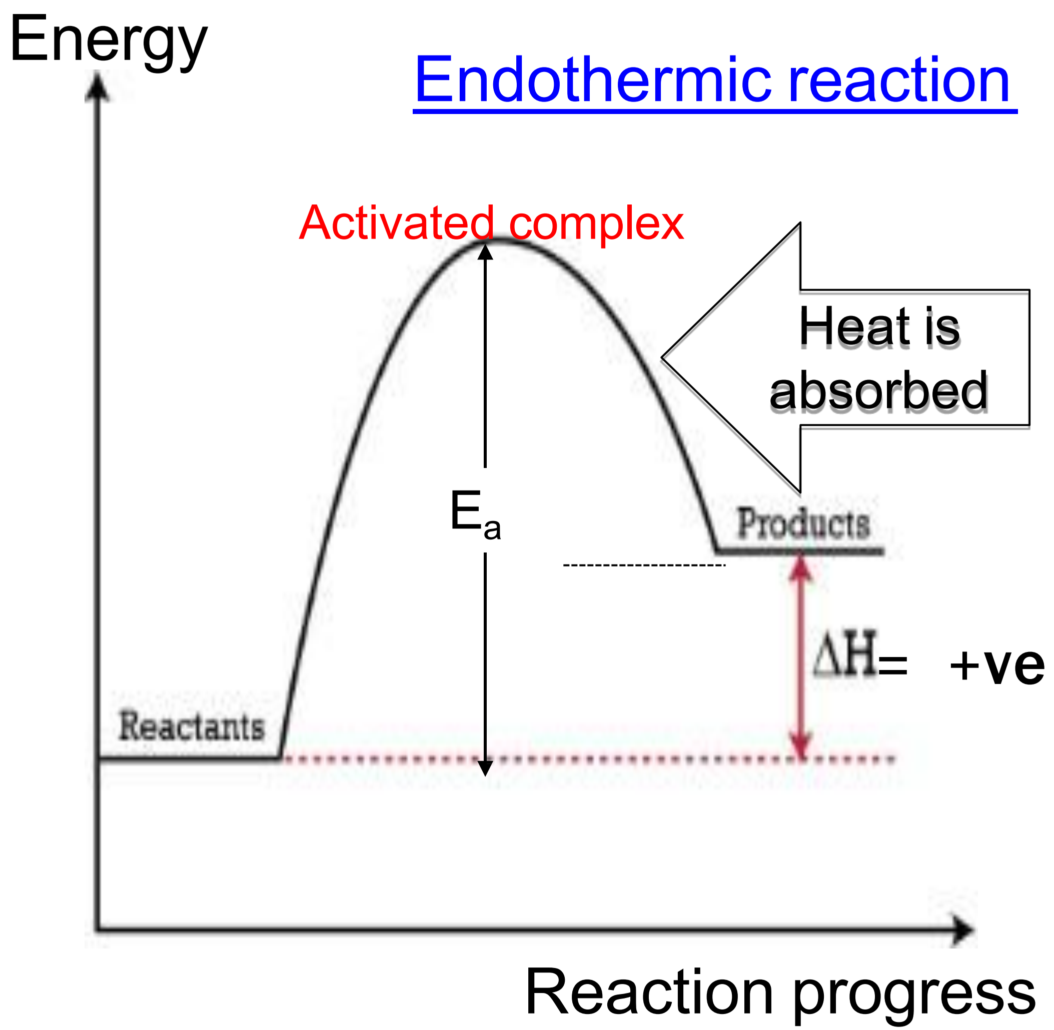
Endothermic reaction

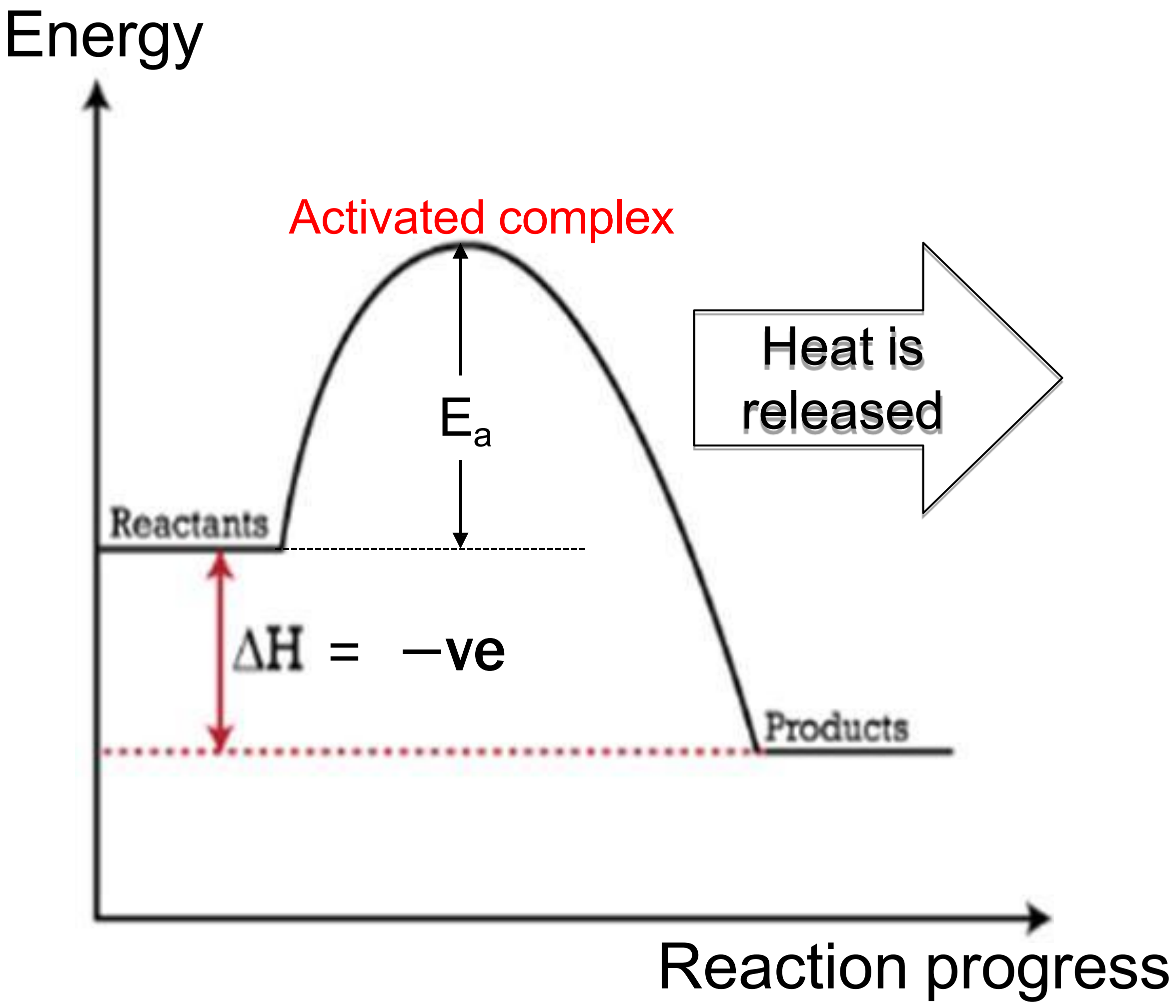


- Heat is **absorbed from** the surroundings.
- Temperature of the surrounding **decreases**.
- $\sum H_{products}$ **>** $\sum H_{reactants}$
- ΔH **> 0** (**positive value**)

Energy Profile Diagram

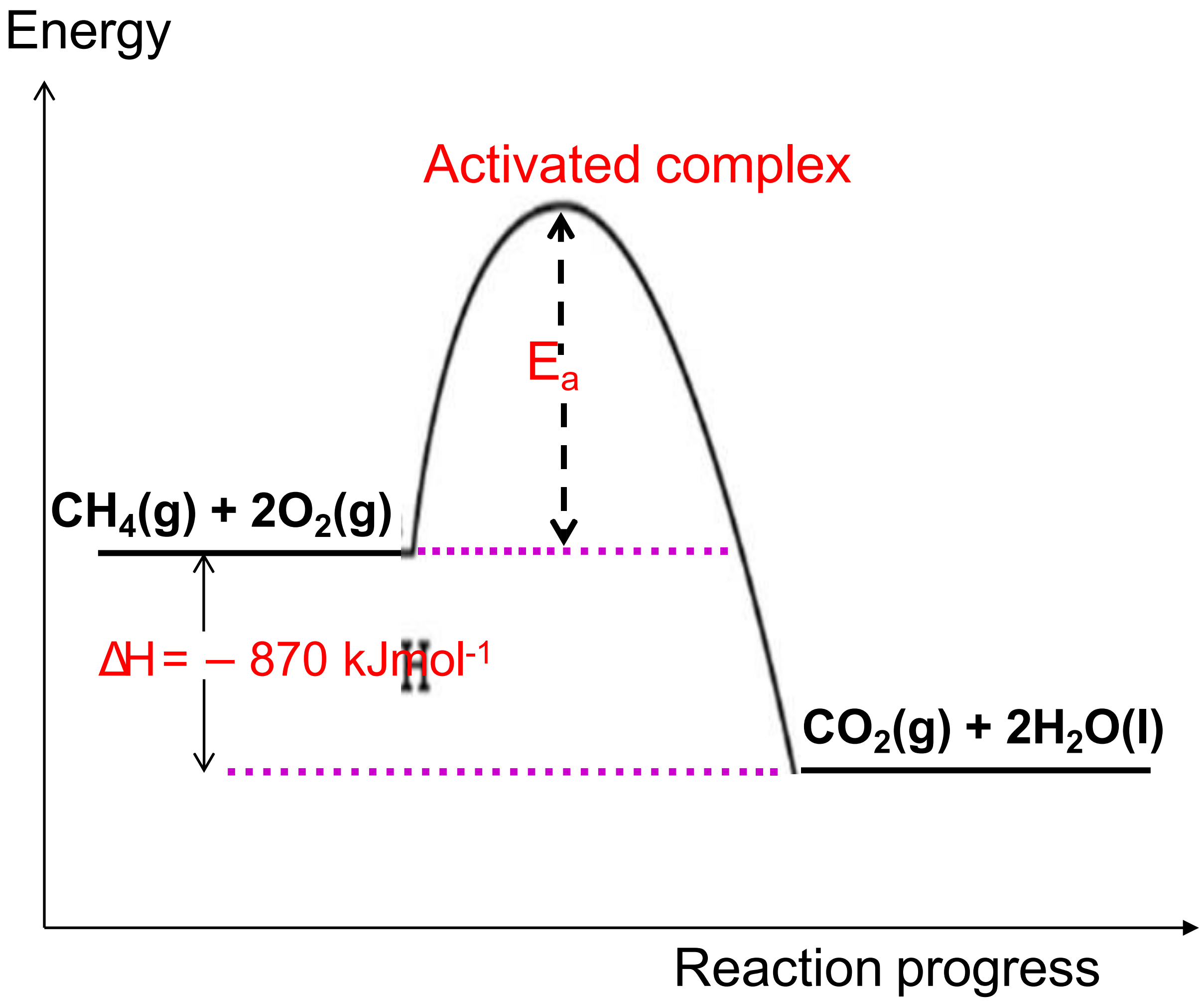
$$\Delta H = \sum H^o_f \text{ products} - \sum H^o_f \text{ reactants}$$





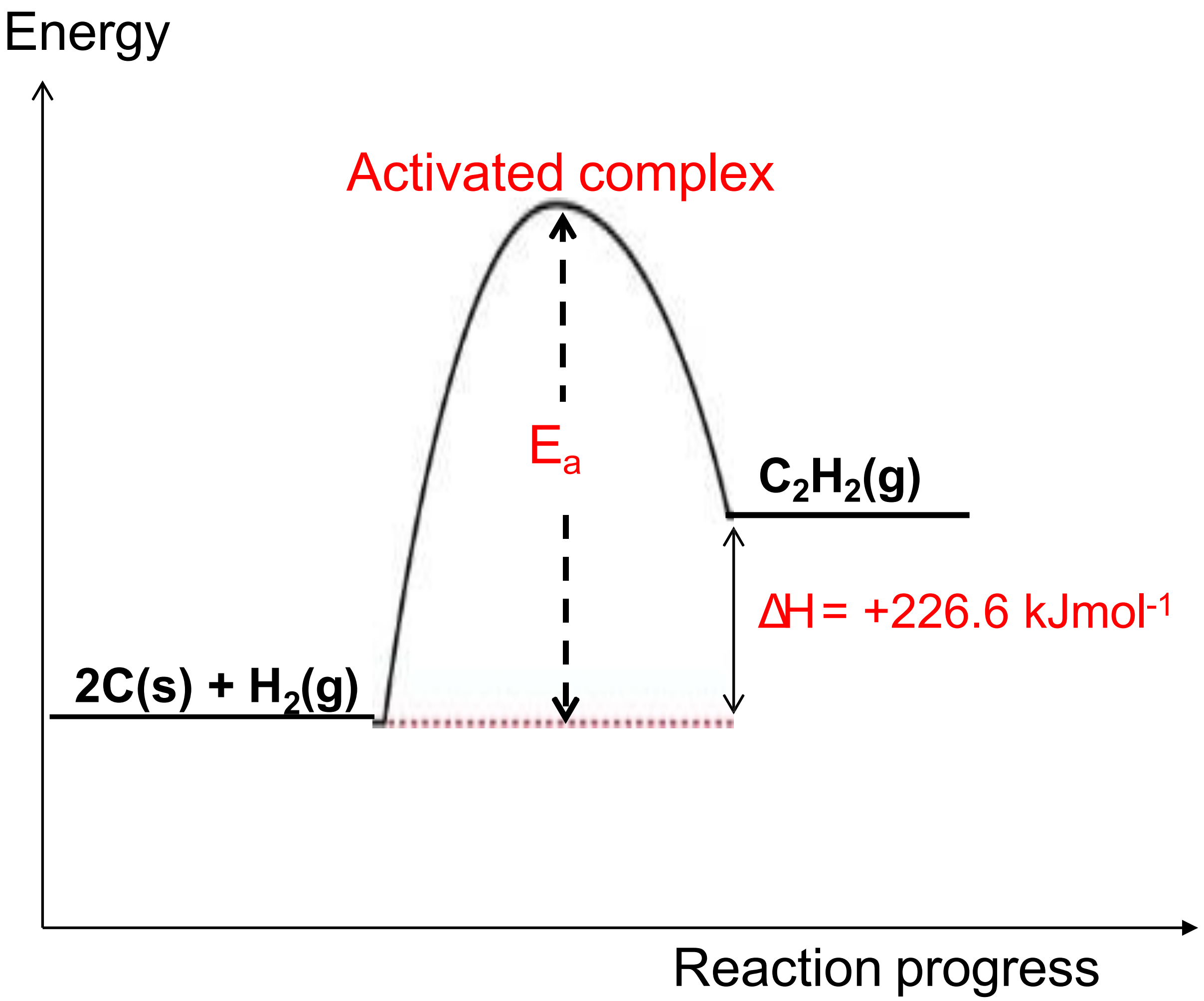
Example of an exothermic reaction

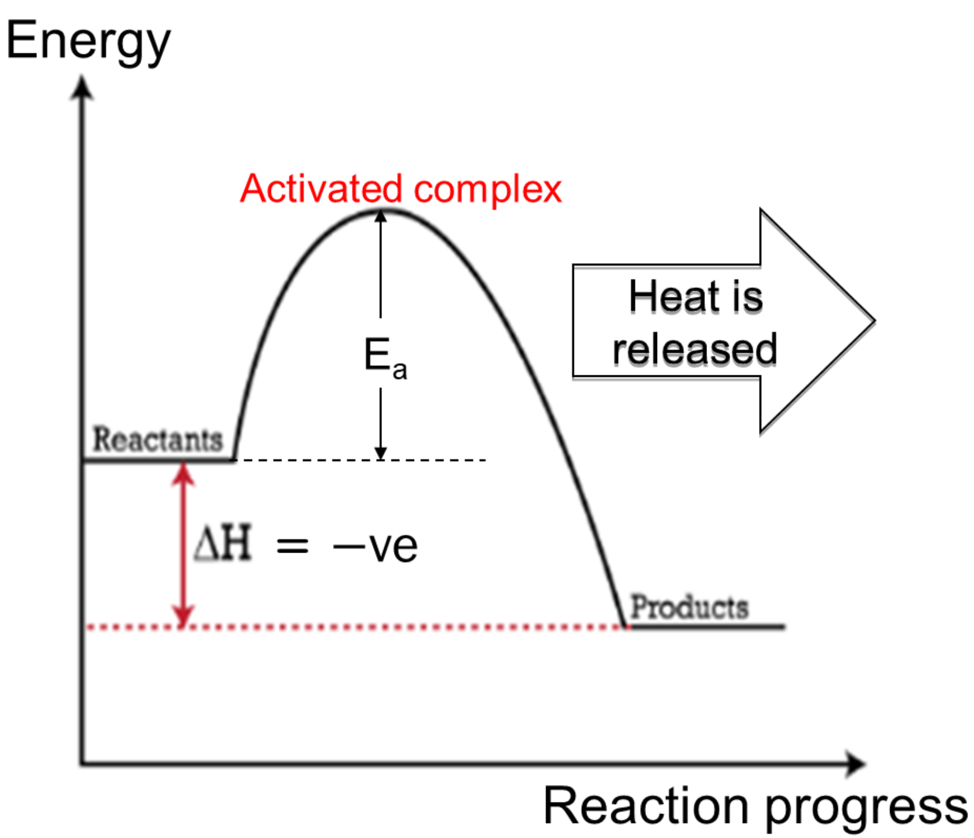
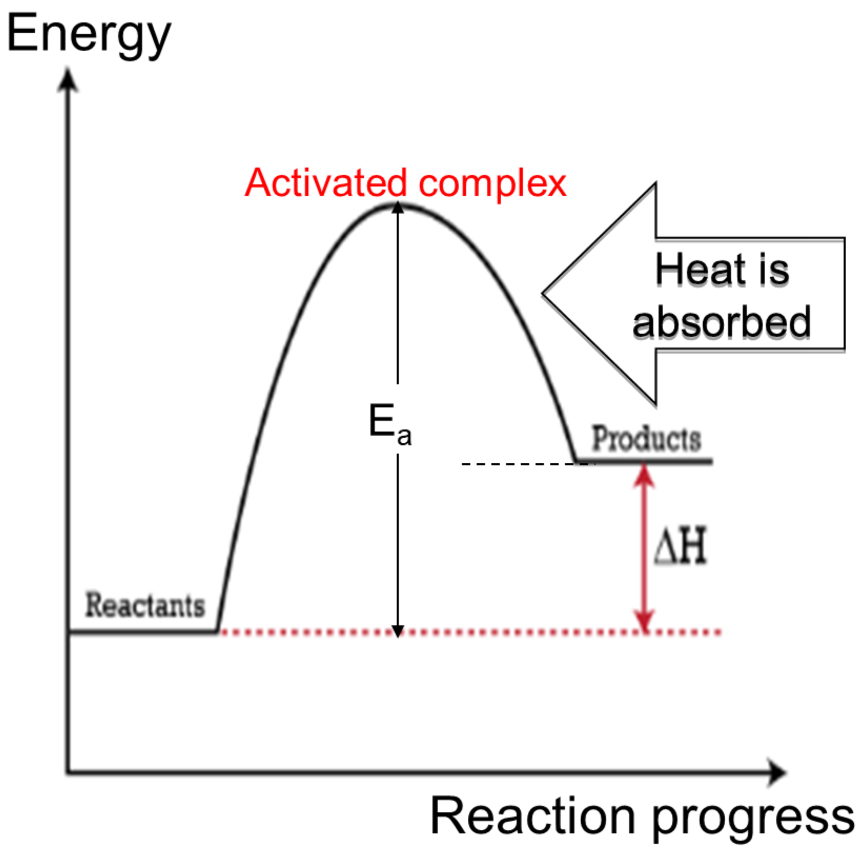
$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$ $\Delta H = -870\ kJmol^{-1}$



Example of an endothermic reaction

$2C(s) + H_2(g) \rightarrow C_2H_2(g)$ $\Delta H = +226.6\ kJmol^{-1}$

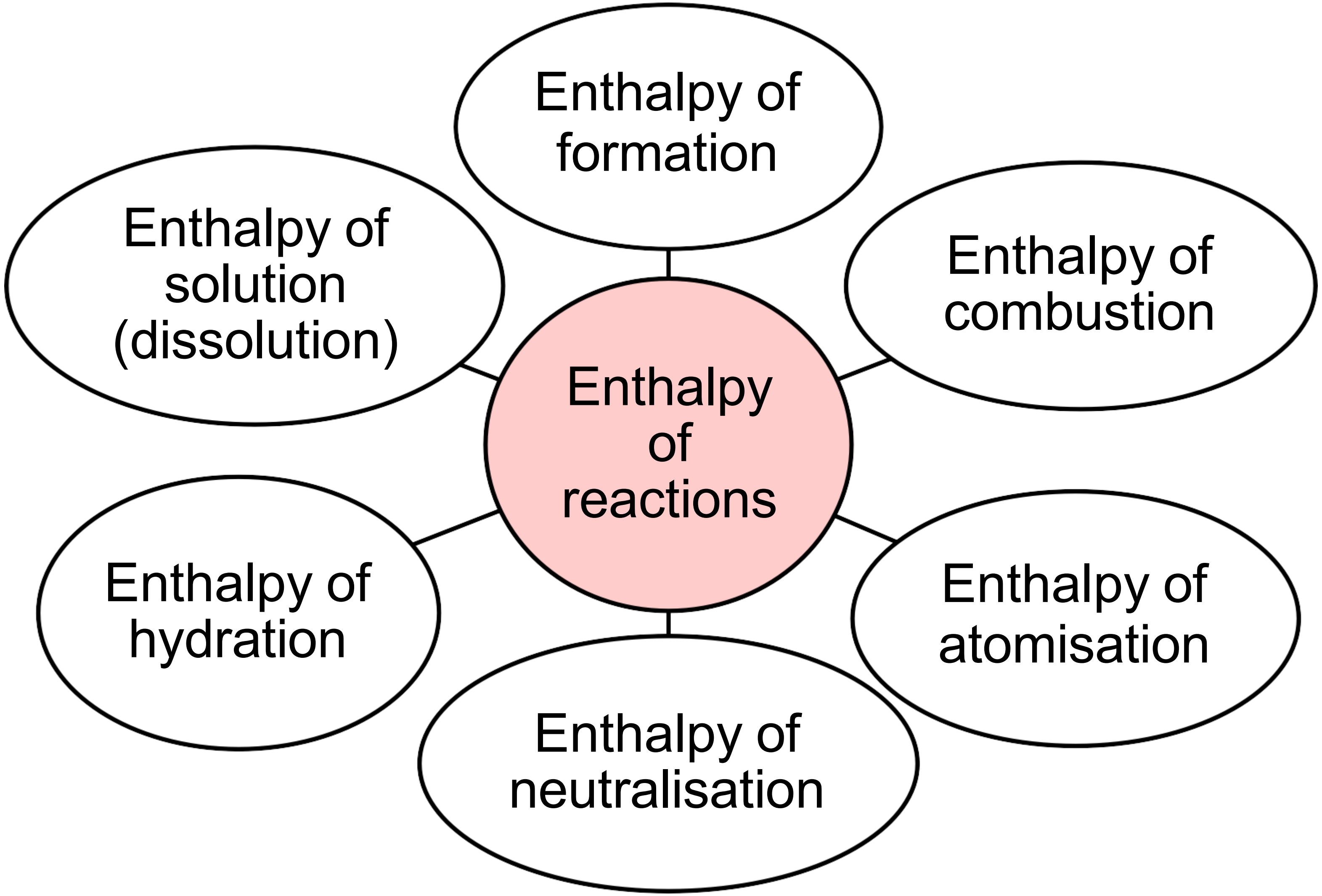


EXOTHERMIC REACTION	ENDOTHERMIC REACTION
	
enthalpy of product is **less** than enthalpy of reactant	enthalpy of products is **more** than enthalpy of reactant
heat is released, **ΔH is negative**	heat is absorbed, **ΔH is positive**.

- The enthalpy change associated with a chemical reaction is known as:

enthalpy of reaction.

Types of Enthalpy of Reactions



The enthalpy of reaction can be represented by **thermochemical equation**.

Thermochemical Equation

- Shows a balanced chemical equation with **physical states of all reactants** and **products** are specified and accompanied with the **value of ΔH**.

$$C(s) + O_2(g) \rightarrow CO_2(g) \qquad \Delta H = -\ 393.5\ kJmol^{-1}$$

- The sign of the enthalpy change is opposite when a thermochemical equation is reversed.

$$CO_2(g \rightarrow C(s) + O_2(g) \qquad \Delta H = +\ 393.5\ kJmol^{-1}$$

- If both sides of an equation are multiplied or divide by a factor *n*, the ΔH must also change by the same factor *n*.

$$2C(s) + 2O_2(g) \rightarrow 2CO_2(g) \qquad \Delta H = 2(-393.5)\ kJ$$

Standard Enthalpy of Formation, $\Delta H^o{}_f$

The **heat change** when one mole of a compound is formed from its elements in their most stable states under standard conditions (P= 1atm and T=25°C).

Examples: $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O\,(l)$ $\Delta H^o = -\,286\ kJmol^{-1}$

$2C(s) + H_2(g) \rightarrow C_2H_2(g)$ $\Delta H^o = +\,237\ kJmol^{-1}$

Exercise:

Write the thermochemical equation for the following enthalpy abbreviations:

$\Delta H^o_f\,[C_6H_6(l)] = +\,64 kJ/mol$ $6C(s) + 3H_2(g) \rightarrow C_6H_6(l)$ $\Delta H^o = +\,64\ kJmol^{-1}$

$\Delta H^o_f\,[Al_2O_3(s)] = -\,1676 kJ/mol$ $2Al(s) + 3/2O_2(g) \rightarrow Al_2O_3(s)$ $\Delta H^o = -\,1676\ kJmol^{-1}$

$\Delta H^o_f\,[C_2H_5OH(l)] = -\,278 kJ/mol$ $2C(s) + 3H_2(g) + \frac{1}{2}O_2(g) \rightarrow C_2H_5OH(l)$ $\Delta H^o = -\,278\ kJmol^{-1}$

- The standard enthalpy of formation of any element in its most stable state is zero.

  Examples:

  $\Delta H^o_f [O_2(g)] = 0$ kJ/mol

  $\Delta H^o_f [C(s)] = 0$ kJmol^{-1}

  $\Delta H^o_f [Cl_2(g)] = 0$ kJ/mol

  $\Delta H^o_f [Na(s)] = 0$ kJmol^{-1}

- The enthalpy of formation can be used to calculate the enthalpy change of a reaction (ΔH).

$$\boxed{\Delta H_{rxn} = \sum m\Delta H^o_f \text{ (products)} - \sum n\Delta H^o_f \text{ (reactants)}}$$

m & n : coefficient from balance equation

Example 1:

By using the data given below, calculate the enthalpy change for the following reactions:

$$C_6H_{12}O_6(s) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l)$$

$\Delta H^o_f\ C_6H_{12}O_6(s) = -\ 1258\ kJmol^{-1}$
$\Delta H^o_f\ H_2O(l) = -\ 286\ kJmol^{-1}\ \Delta$
$H^o_f\ CO_2(g) = -\ 392\ kJmol^{-1}$

Solution:

$$\boxed{\Delta H_{rxn} = \sum m\Delta H^o_f\ (products) - \sum n\Delta H^o_f\ (reactants)}$$

$$\Delta H_{rxn} = [6(\Delta H^o_f\ CO_2) + 6(\Delta H^o_f\ H_2O)] - [\Delta H^o_f\ C_6H_{12}O_6 + 6(\Delta H^o_f\ O_2)]$$

$$= [6(-\ 392) + 6(-\ 286)] - [-\ 1258 + 6(0)]$$

$$= -\ 2810\ kJ$$

$$\Delta\ H^o_{rxn} = -\ 2810\ kJmol^{-1}$$

Example 2:

A thermochemical equation is given below:

$$C(s) + O_2(g) \rightarrow CO_2(g) \qquad \Delta H = -393\ kJmol^{-1}$$

a) Name two possible enthalpies represented by the equation.
b) What is the heat change if 1 g of carbon is reacted with excess O_2?
c) What is the heat change if 1 g of carbon dioxide formed in the reaction?

Solution:

a) Enthalpy of formation of CO_2 (g)
Enthalpy of combustion of C (s)

b) $1\ mol \equiv 12\ g\ C \equiv -396\ kJ$

$$\therefore 1\ g\ C \equiv \frac{1}{12} \times (-396) = \mathbf{-33\ kJ}$$

c) $1\ mol \equiv 44\ g\ CO_2 \equiv -396kJ$

$$\therefore 1g\ CO_2 \equiv \frac{1}{44} \times (-396)$$

$$= \mathbf{-9\ kJ}$$

Standard Enthalpy of Combustion, $\Delta H^o{}_c$

The **heat released** when one mole of a substance is completely burned in excess oxygen under standard conditions (P= 1 atm and T=25°C).

Examples:

$$C(s) + O_2(g) \rightarrow CO_2(g) \qquad \Delta H^o = -\ 393.5\ kJmol^{-1}$$

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g) \qquad \Delta H^o = -\ 892\ kJmol^{-1}$$

Example 1: Write the thermochemical equation for the combustion of heptane, C_7H_{16} and benzene, C_6H_6 if the enthalpies of combustion are $-$ 4803 $kJmol^{-1}$ and $-$1368 $kJmol^{-1}$, respectively.

Solution:

$$C_7H_{16}(l) + 11O_2(g) \rightarrow 7CO_2(g) + 8H_2O(l) \qquad \Delta H^o = -\ 4803\ kJmol^{-1}$$

$$C_6H_6(l) + \frac{15}{2}O_2(g) \rightarrow 6CO_2(g) + 3H_2O(l) \qquad \Delta H^o = -\ 1368\ kJmol^{-1}$$

Standard Enthalpy of Atomisation, ΔH°_{atom}

The **heat absorbed** when one mole of gaseous atom is formed from its element in its most stable state under standard conditions (P= 1atm and T=25^{o}C).

Examples: $Na(s) \rightarrow Na(g)$ $\Delta H^{o} = +\ 109\ kJmol^{-1}$

$\frac{1}{2}\ Cl_2(g) \rightarrow Cl(g)$ $\Delta H^{o} = +\ 123\ kJmol^{-1}$

- The enthalpy of atomisation of noble gases is **zero**, as they already exist as monoatomic gases under standard conditions.

Example:

Write the thermochemical equations for the atomisation of fluorine, nitrogen and iron if the heat changes are +79.1, +472.8 and +418 $kJmol^{-1}$, respectively.

Solution:

$\frac{1}{2} F_2(g) \rightarrow F(g) \qquad \Delta H^o = + 79.1\ kJmol^{-1}$

$\frac{1}{2} N_2(g) \rightarrow N(g) \qquad \Delta H^o = + 472.8\ kJmol^{-1}$

$Fe(s) \rightarrow Fe(g) \qquad \Delta H^o = + 418\ kJmol^{-1}$

Standard Enthalpy of Neutralisation, $\Delta H^o_{neutralisation}$

The **heat released** when one mole of water, H_2O is formed from the neutralisation of acid and base under standard conditions (P= 1 atm and T=25^{o}C).

Examples:

$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(aq)$ $\Delta H^o = -58\ kJmol^{-1}$

$CH_3COOH(aq) + NaOH(aq) \rightarrow CH_3COONa(aq) + H_2O(aq)$ $\Delta H^o = -56.1\ kJmol^{-1}$

- The neutralisation of strong base and strong acid releases $\approx -57.3\ kJmol^{-1}$
- The standard enthalpy change involving weak acid or bases $< -57.3\ kJmol^{-1}$

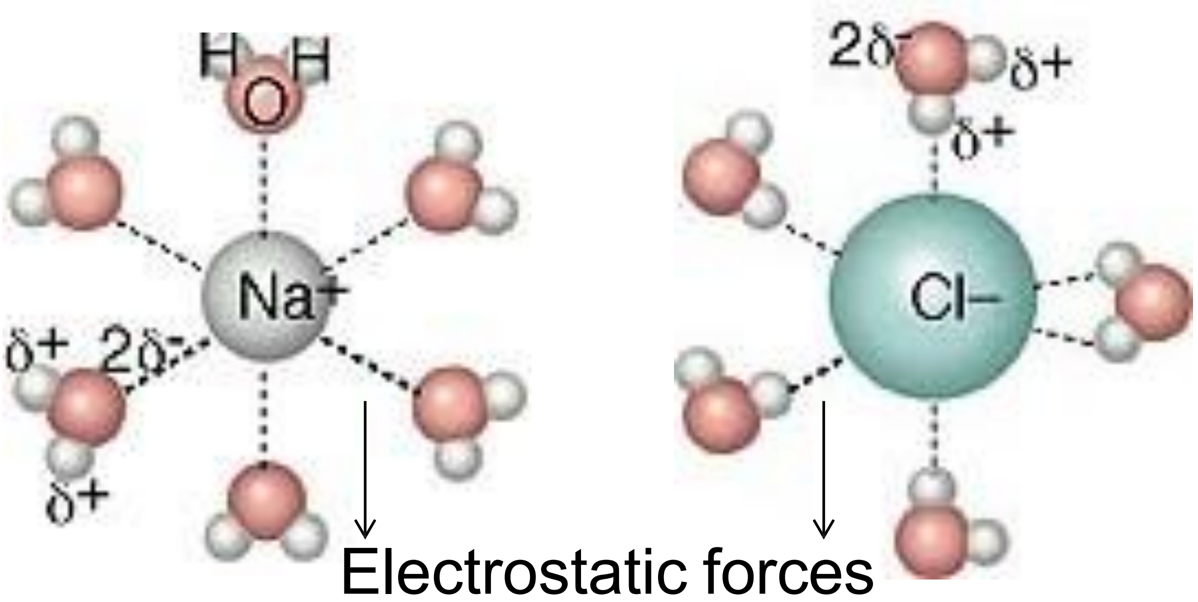
Standard Enthalpy of Hydration, ΔH^{o}_{hyd}

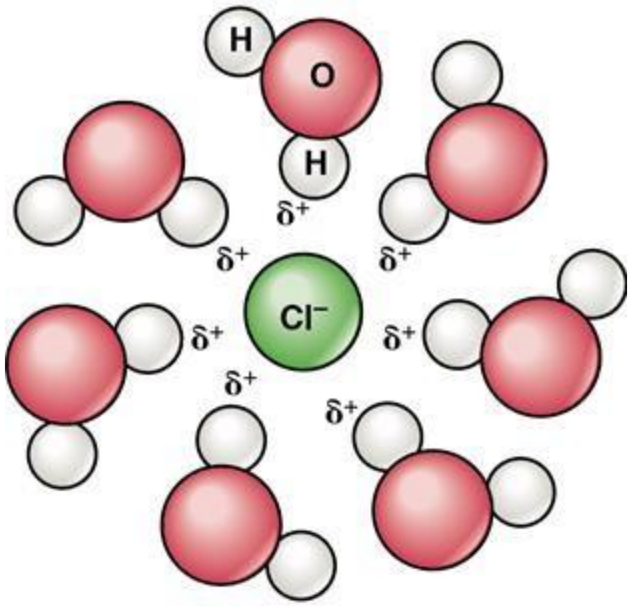
The **heat released** when one mole of gaseous ion is dissolved in water to form one mole of hydrated ion under standard conditions (P= 1 atm and T=25°C).

Examples: $Na^{+}(g) \rightarrow Na^{+}(aq)$ $\Delta H = -406\ kJmol^{-1}$

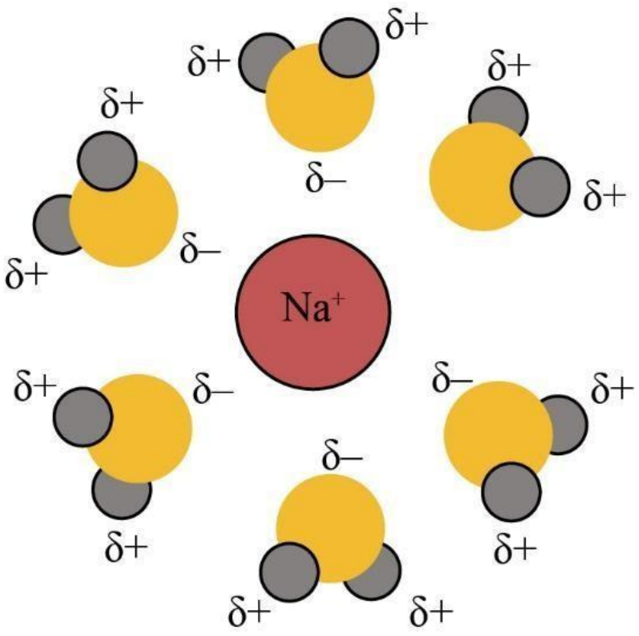
$Cl^{-}(g) \rightarrow Cl^{-}(aq)$ $\Delta H = -363\ kJmol^{-1}$

Hydration is a process in which water molecules surround positive and negative ions.





Hydrated Cl^{-} ion



Hydrated Na^{+} ion

- The partial positive end of the water molecules is attracted to the anion, and the partial negative end of the water molecules is attracted to the cation through **electrostatic forces**.
- Ion-solvent bonds are formed.
- Energy is released when ions become hydrated.

Standard Enthalpy of Solution, ΔH^o_{soln} (or Dissolution)

The **heat change** when one mole of a substance dissolves completely in water to form a dilute solution under standard conditions (P= 1 atm and T=25ºC).

Example:

$$KCl(s) \rightarrow K^+(aq) + Cl^-(aq) \qquad \Delta H = + 690 \ kJmol^{-1}$$

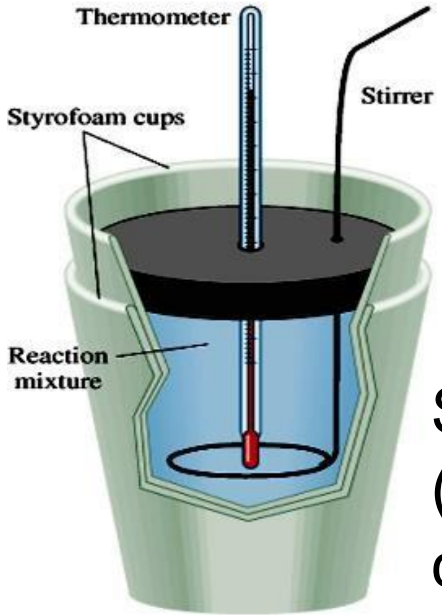
Note:

If the solvent is not water, this is referred to as the enthalpy of solvation.

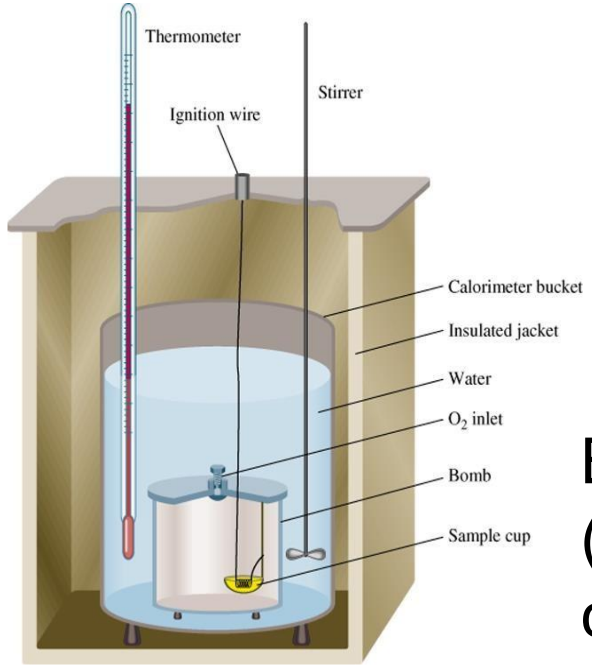
2.2 CALORIMETRY

Calorimetry

- A method used in the laboratory to **measure the heat change of a reaction**.
- Apparatus used is known as the calorimeter.
- Examples of calorimeter :



Simple calorimeter (measured at constant pressure)



Bomb calorimeter (measured at constant volume)

- The general principle of a calorimeter:

Heat released by a chemical reaction = Heat absorbed by the surrounding

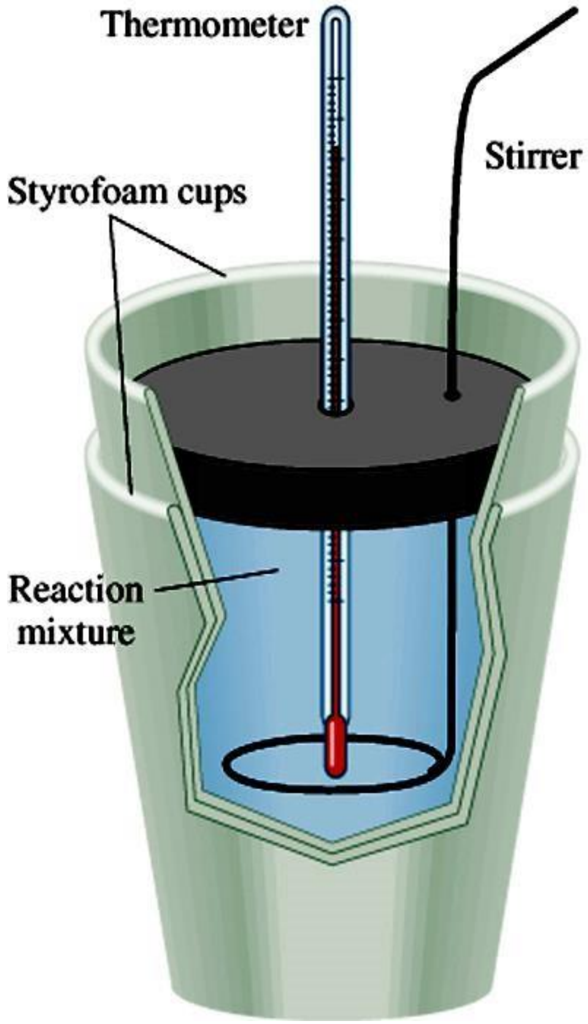
- Heat released or absorbed is given by:

$\boldsymbol{q = mc\Delta T}$ @ $\boldsymbol{q = C\Delta T}$

where
m = mass (in g)
c = specific heat capacity of a substance (J g^{-1}°C^{-1})
C = heat capacity (J°C^{-1})
ΔT = temperature change (°C)

- **Specific heat capacity, c**
  - is the amount of heat required to raise the temperature of **one gram of the substance** by one degree Celsius (Jg^{-1}°C^{-1}).
- **Heat capacity, C**
  - is the amount of heat required to raise the temperature of a **given quantity of the substance** by one degree Celsius (J°C^{-1}).

<u>In a simple calorimeter</u>:



- The outer styrofoam cup insulate the reaction mixture from the surroundings

  (it is assumed that no heat is lost to the surroundings).

- **Heat released by the reaction is absorbed by the solution**.

- Thus,

$$\boxed{-q_{rxn} = (mc\Delta T)_{solution} + (C\Delta T)_{cal}}$$

Example:

When a student mixes 50.00 mL of 1.0 M HCl and 50.00 mL of 1.0 M NaOH in a coffee-cup calorimeter, the temperature of the resultant solution increases from 21.0^oC to 27.5^oC. Calculate the enthalpy of neutralisation of the reaction. Assume that heat absorbed by calorimeter is negligible. The specific heat capacity of the solution is 4.18 J g^o C^{-1} and density of the solution is1.0 g mL^{-1}.

Solution:

$$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$$

$$-q_{rxn} = (mc\Delta T)_{solution} + (C\Delta T)_{cal}$$

since the heat absorbed by calorimeter is negligible,

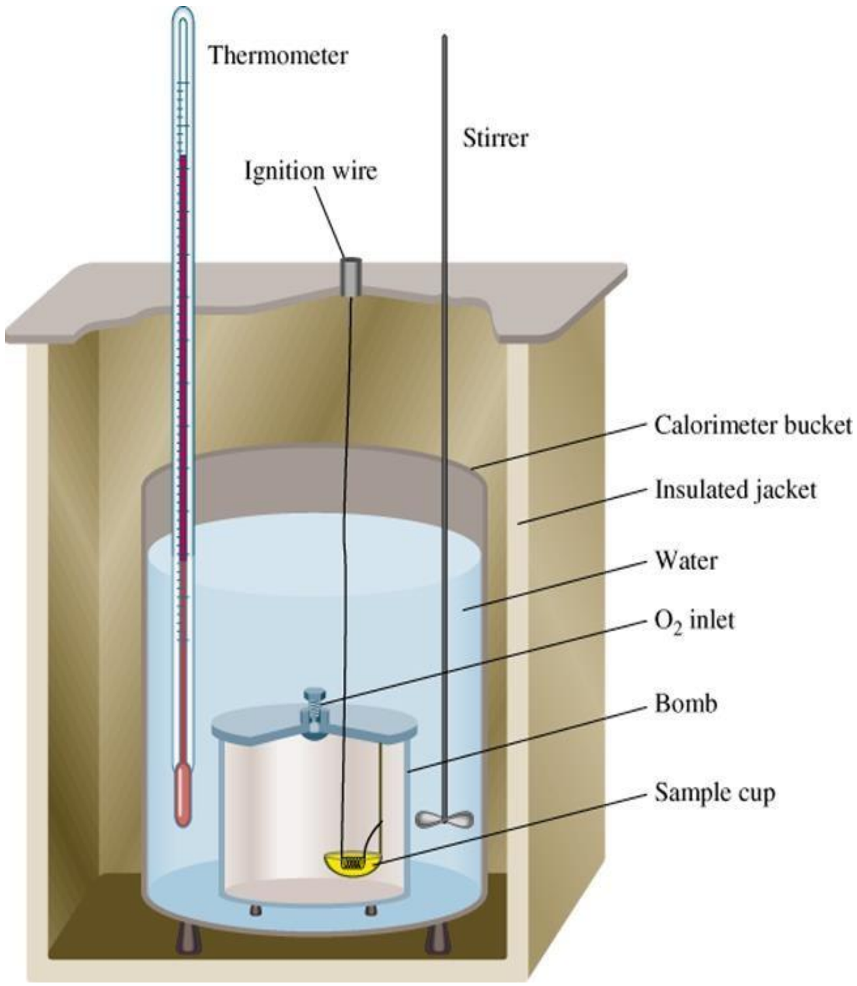
$$-q_{rxn} = (mc\Delta T)_{solution}$$
$$= (100 \times 4.18 \times 6.5)$$
$$= 2717 \text{ J}$$
$$= 2.717 \text{ kJ}$$

$$n_{HCl} = n_{NaOH} = MV$$
$$= 1.0 \times 50 \times 10^{-3}$$
$$= 0.05 \text{ mol}$$
$$n_{H_2O} = 0.05 \text{ mol}$$

$$0.05 \text{ mol } H_2O = -2.717 \text{ kJ}$$
$$1 \text{ mol } H_2O = -54.34 \text{ kJ}$$

enthalpy of neutralisation = -54.34 kJ mol^{-1}

- <u>In a bomb calorimeter</u>:



- Bomb calorimeter is made of a strong steel vessel filled with pure O_2 at a pressure of 25 atm and immersed in an exactly **KNOWN** mass of water.

- The **heat released by the reaction is absorbed by both the water and the metal parts of the calorimeter**.

- Thus,

$$\boxed{-q_{rxn} = (mc\Delta T)_{water} + (C\Delta T)_{cal}}$$

Example: 2.00 g benzoic acid, C_6H_5COOH was burned in a bomb calorimeter containing 1.00 kg water. The temperature of the water increased by 5.26 ^oC. Calculate the **enthalpy of combustion** of benzoic acid if the heat capacity of the calorimeter is 5.70 x 10^3 J^{o}C^{-1}.

Solution:

$$C_6H_5COOH(s) + \frac{15}{2} O_2(g) \rightarrow 7CO_2(g) + 3H_2O(l)$$

$$-q_{rxn} = (mc\Delta T)_{water} + (C\Delta T)_{cal}$$

$$= (1000 \text{ g} \times 4.18 \text{ Jg}^{-1\,o}C^{-1} \times 5.26\ ^oC) + (5.70\times10^3 \text{ J}^oC^{-1} \times 5.26\ ^oC)$$

$$= 51968.8 \text{ J} = 5.2 \times 10^4 \text{ J}$$

$$n_{benzoic\ acid} = \frac{2}{122} = 0.0164 \text{ mol}$$

$$0.0164 \text{ mol } C_6H_5COOH \equiv -5.2 \times 10^4 \text{ J}$$

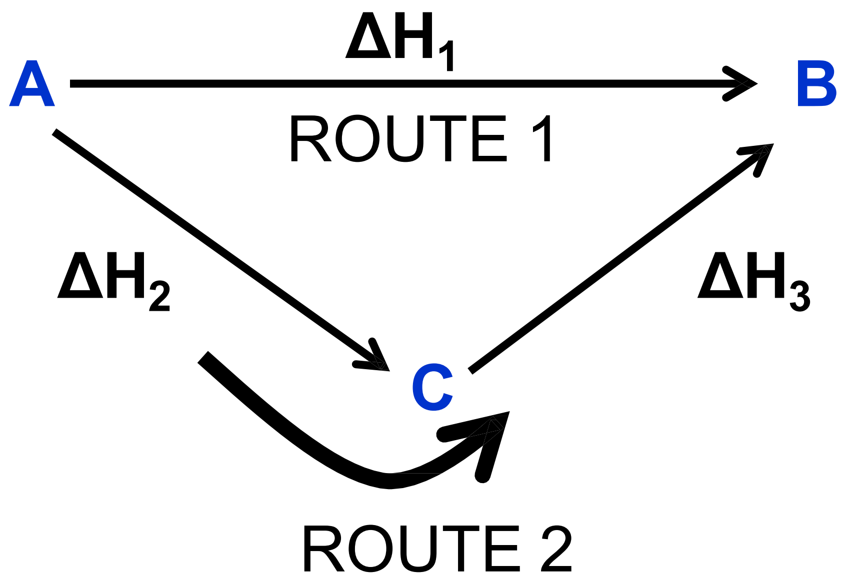
$$\therefore 1 \text{ mol } C_6H_5COOH \equiv -3.17 \times 10^6 \text{ J}$$

$$\Delta \mathbf{H_{combustion} = -3170.71\ kJmol^{-1}}$$

2.3
HESS’S LAW

Hess's Law

- **Hess's law states that** when reactants are converted to products, the change in enthalpy is the same whether the reaction takes place in one step or in a series of steps.



- According to Hess's law:

$$\boxed{\Delta H_1 = \Delta H_2 + \Delta H_3}$$

- Enthalpy changes can be calculated by:
  i) Algebraic method
  ii) Energy cycle method

ALGEBRAIC METHOD

Step 1

Write thermochemical equations for the given enthalpy change.

Step 2

Write the expected thermochemical equation.

Step 3

Adjust the thermochemical equations so that they add up to the expected equation.

- You can reverse the equations. When you do this, the ΔH changes its sign.
- You can multiply the equations by any common factor. When you do this, you must also multiply the ΔH by the same factor.

Step 4

Add up the ΔH of all the adjusted thermochemical equations.

ENERGY CYCLE METHOD

Draw the energy cycle and apply Hess's Law to calculate the unknown value of ΔH.

Example: Find the standard enthalpy of formation of ethane, C_2H_6 (g).
Given,

$\Delta H°_c$ [C(s)] = - 393 $kJmol^{-1}$

$\Delta H°_c$ [H_2(g)] = - 286 $kJmol^{-1}$

$\Delta H°_c$ [C_2H_6(g)) = - 1560 $kJmol^{-1}$

Solution: Method (1) Algebraic method

Step 1: Write the thermochemical equations for the given change in enthalpies.

$C(s) + O_2(g) \rightarrow CO_2(g)$ $\Delta H°_c = -393\ kJmol^{-1}$

$H_2(g) + ½\ O_2(g) \rightarrow H_2O(g)$ $\Delta H°_c = -286\ kJmol^{-1}$

$C_2H_6(g) + 7/2\ O_2(g) \rightarrow 2CO_2(g) + 3\ H_2O(g)$ $\Delta H°_c = -1560\ kJmol^{-1}$

Step 2: Write the expected chemical equation for the formation of ethane:

$$2C(s) + 3H_2(g) \rightarrow C_2H_6(g)$$

Step 3: Adjust the thermochemical equations :

$$2CO_2(g) + 3H_2O(g) \rightarrow C_2H_6(g) + \frac{7}{2}O_2 \qquad \Delta H = +\ 1560\ kJ$$

$$2C\ (s) + 2O_2(g) \rightarrow 2CO_2\ (g) \qquad \Delta H = 2(-\ 393)\ kJ$$

$$3H_2\ (g) + \frac{3}{2}O_2(g) \rightarrow 3H_2O\ (g) \qquad \Delta H = 3(-\ 286)\ kJ$$

$$2C\ (s) + 3H_2(g) \rightarrow C_2H_6\ (g)$$

<u>Step 3</u>: Calculating the standard enthalpy formation of ethane:

$$2CO_2(g) + 3H_2O(g) \rightarrow C_2H_6(g) + \frac{7}{2}O_2 \qquad \Delta H = +1560\ kJ$$

$$2C(s) + 2O_2(g) \rightarrow 2CO_2(g) \qquad \Delta H = 2(-393)\ kJ$$

$$3H_2(g) + \frac{3}{2}O_2(g) \rightarrow 3H_2O(g) \qquad \Delta H = 3(-286)\ kJ$$

$$2C(s) + 3H_2(g) \rightarrow C_2H_6(g)$$

$$\Delta H = (+1560 + (-786) + (-858)$$

$$= -84$$

$$\boldsymbol{\Delta H^{o}_{formation} = -84\ kJmol^{-1}}$$

Method (2) Energy cycle method

$$
\begin{tikzcd}
2C(s) \arrow[d, "+2O_2" near start, "2\Delta H_c\,[C(s)]"'] & + & 3H_2(g) \arrow[r, "\Delta H_f"] \arrow[d, "+\frac{3}{2}O_2" near start, "3\Delta H_c\,[H_2(g)]"'] & C_2H_6(g) \arrow[dl, "+\frac{7}{2}O_2" near start, "\Delta H_c\,[C_2H_6(g)]"] \\
2CO_2(g) & + & 3H_2O(g) &
\end{tikzcd}
$$

$$\Delta H_f + \Delta H_c\, C_2H_6(g) = 2\Delta H_c\, C(s) + 3\Delta H_c\, H_2(g)$$

$$\Delta H_f + (-1560) = (-786) + (-858)$$

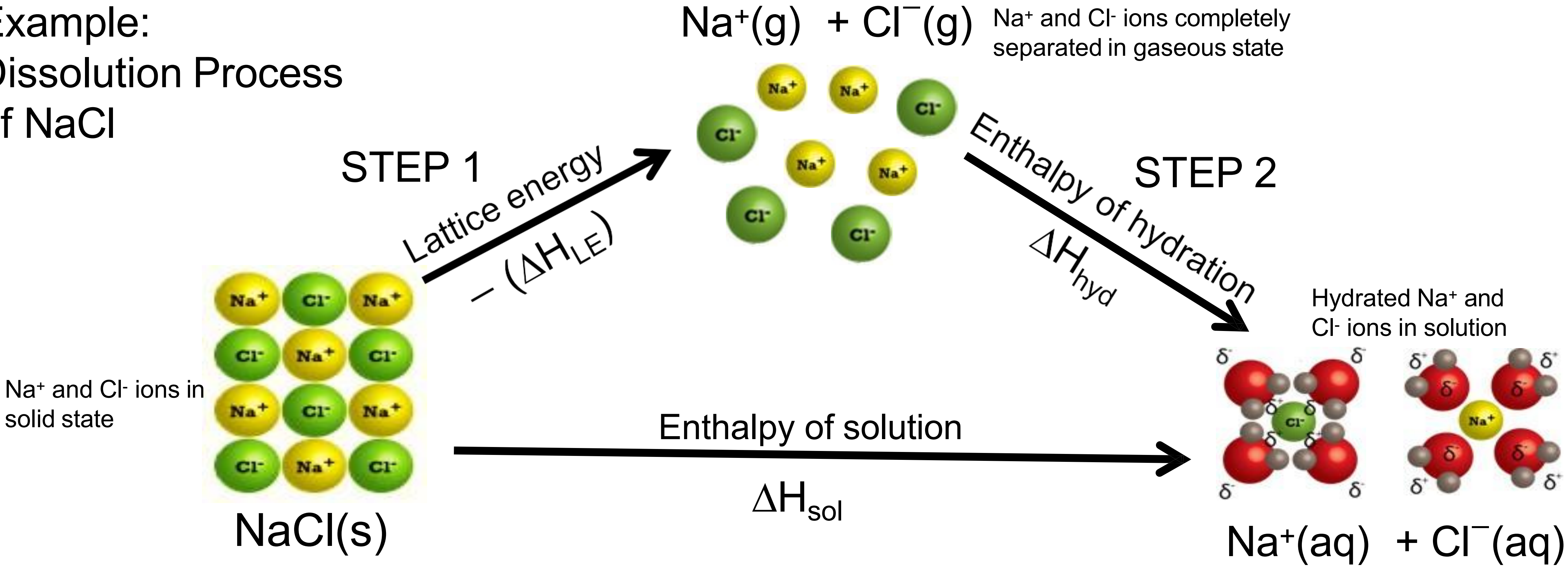
$$\Delta H_f = -84$$

$$\mathbf{\Delta H^o_{formation} = -84\ kJmol^{-1}}$$

DISSOLUTION PROCESS

- When an **ionic solid is dissolved in water**, two stages are involved in the solution process:
  1. Breaking up the ionic lattice to form gaseous ions (energy is absorbed).
  2. Hydration of the gaseous ions.

Example:
Dissolution Process
of NaCl



Example:

Calculate the enthalpy of solution of KCl if its lattice energy is $-690\ kJmol^{-1}$ and the enthalpy of hydration is $-672\ kJmol^{-1}$.

Solution:

$$\begin{tikzcd}
KCl(s) \arrow[r, "\Delta H_{soln}"] \arrow[dr, "-(\Delta H_{lattice})"'] & K^+(aq) + Cl^-(aq) \\
 & K^+(g) + Cl^-(g) \arrow[u, "\Delta H_{hyd}"']
\end{tikzcd}$$

$$\Delta H_{soln} = -(\Delta H_{lattice}) + \Delta H_{\ hydrate}$$

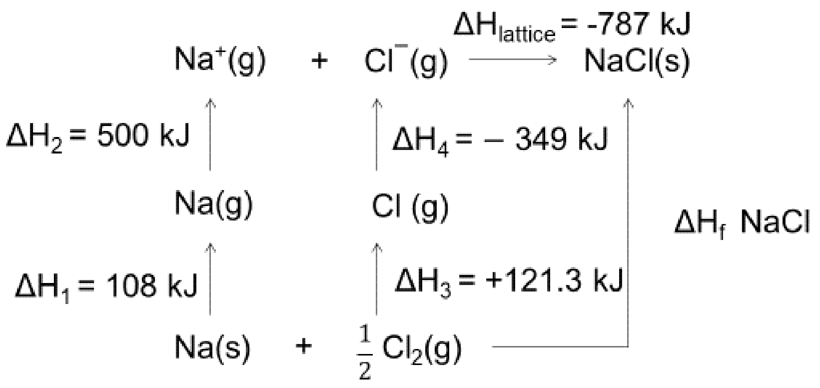
$$= -(-690) + (-672)\ kJ$$

$$= \mathbf{+18\ kJmol^{-1}}$$

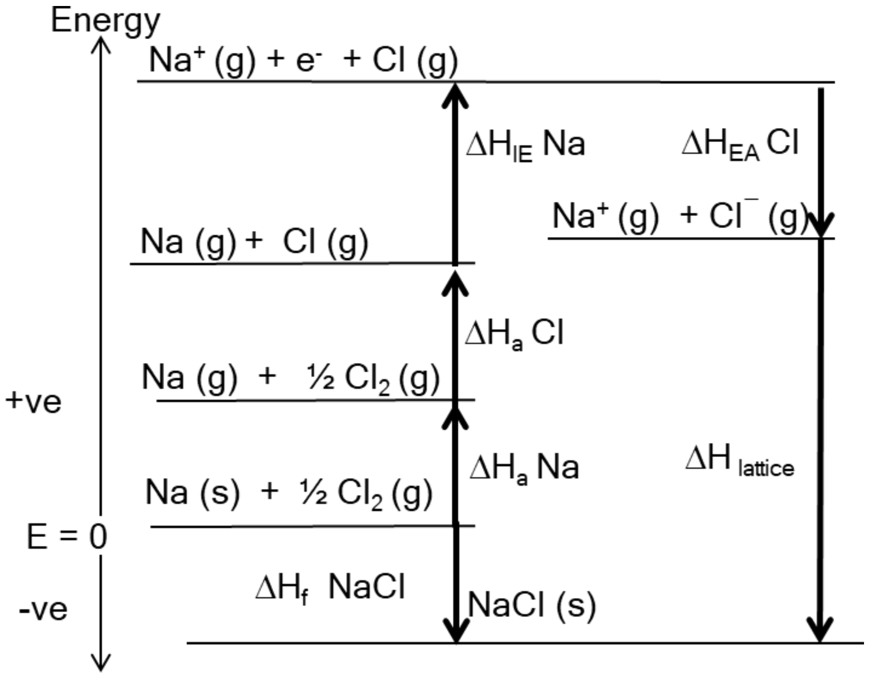
2.4 BORN-HABER CYCLE

BORN-HABER CYCLE

- Born-Haber cycle is an energy cycle that shows the related enthalpy changes in the formation of an ionic compound.
- A Born-Haber cycle is useful for determining **lattice energy**, which cannot be obtained directly from an experiment.
- A Born-Haber cycle could be thought of as a complex Hess's law rectangle.



- A Born-Haber cycle is sometimes drawn in such a way that the relative length of the arrows is proportional to the energy change involved.
- Moreover, endothermic reactions are drawn with arrows pointing up the page, and exothermic reactions are drawn with arrows pointing down the page.



LATTICE ENERGY

The **heat released** when one mole of an ionic solid is formed from its constituent gaseous ions.

Examples:

$$Na^+(g) + Cl^-(g) \rightarrow NaCl(s) \quad \Delta H = -787\ kJmol^{-1}$$

$$Ca^{2+}(g) + 2F^-(g) \rightarrow CaF_2(s) \quad \Delta H = -2600\ kJmol^{-1}$$

Note: When an **ionic solid is broken** into its constituent gaseous ions, **energy is absorbed**.

$$NaCl(s) \rightarrow Na^+(aq) + Cl^-(aq) \qquad \Delta H = +787\ kJmol^{-1}$$

- Lattice energy measures the strength of the ionic bonds in an ionic solid.

- The stronger the ionic bond, the more negative the lattice energy.

- Factors affecting the lattice energy are:
  1) Ionic charges
  2) Ionic radii

- The lattice energy becomes more negative (more heat is released).
  1) the ionic charges increase, the greater the ionic charges, the stronger the electrostatic attraction between ions, the stronger the ionic bond.
  2) The smaller the ionic radii, the stronger the electrostatic attraction between ions, the stronger the ionic bond.

Example 1:

Given:

$Na^+ (g) + Cl^-(g) \rightarrow NaCl(s) \quad \Delta H_{lattice} = -787\ kJmol^{-1}$

$Na^+ (g) + Br^-(g) \rightarrow NaBr(s) \quad \Delta H_{lattice} = -733\ kJmol^{-1}$

Would the ionic bond be stronger or weaker for NaCl than it is for NaBr? Explain your answer.

Solution:

Both ionic solids have the same charge. However, the lattice energy of NaCl is more negative than NaBr because the ionic radius of Cl^- is smaller than Br^-. Thus, the electrostatic attraction between Na^+ and Cl^- are stronger compared between Na^+ and Br^-. Therefore, NaCl has stronger ionic bond than NaBr.

Example 2:

Given:

$Na^+ (g) + Cl^-(g) \rightarrow NaCl(s) \qquad \Delta H_{lattice} = -787\ kJmol^{-1}$

$Mg^{2+} (g) + 2Cl^-(g) \rightarrow MgCl_2(s) \quad \Delta H_{lattice} = -2489\ kJmol^{-1}$

$MgCl_2$ has more negative lattice energy value than NaCl. Explain.

Solution:

The lattice energy of $MgCl_2$ is more negative than NaCl because:

i. Mg^{2+} has a higher ionic charge than Na^+

ii. the ionic radius of Mg^{2+} is smaller than Na^+

Thus, $MgCl_2$ has stronger ionic bond than NaBr. The stronger the ionic bond, the more negative the lattice energy value.

In constructing a Born-Haber cycle, the following information are required:

IONISATION ENERGY

The minimum energy required to remove one mole of electrons from one mole of gaseous atoms or gaseous positive ions.

Examples:

$Mg(g) \rightarrow Mg^{+}(g) + e$ $\quad IE_1 = +740\ kJmol^{-1}$ — 1st ionisation energy

$Mg^{+}(g) \rightarrow Mg^{2+}(g) + e$ $\quad IE_2 = +1400\ kJmol^{-1}$ — 2nd ionisation energy

ELECTRON AFFINITY

The heat change when one mole of electrons is added to one mole of gaseous atoms or gaseous ions.

Examples:

$O(g) + e \rightarrow O^{-}(g)$ $\quad EA_1 = -141.4\ kJmol^{-1}$ — 1st electron affinity

$O^{-}(g) + e \rightarrow O^{2-}(g)$ $\quad EA_2 = +790.8\ kJmol^{-1}$ — 2nd electron affinity

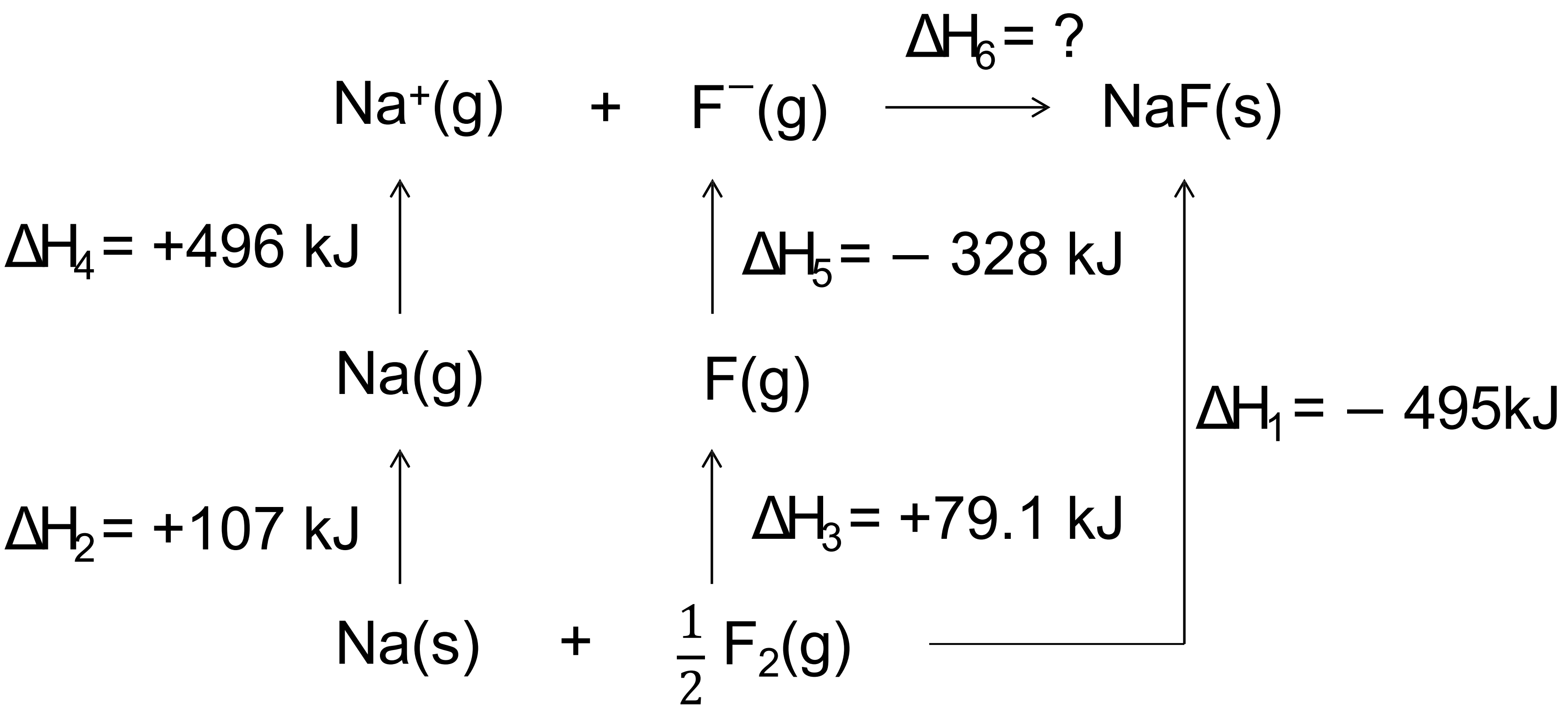
Example 1:

Construct a Born-Haber cycle from the data below to determine the lattice energy of **NaF**.

Reaction	**ΔH (kJ mol^{-1})**
$Na(s) \rightarrow Na(g)$	+107
$Na(g) \rightarrow Na^+(g) + e^-$	+496
$Na(s) + ½F_2(g) \rightarrow NaF(s)$	– 569
$½ F_2(g) \rightarrow F(g)$	+79.1
$F(g) + e^- \rightarrow F^-(g)$	– 328
$Na^+(g) + F^-(g) \rightarrow NaF(s)$	?

Solution 1:

<u>Using energy cycle diagram</u>



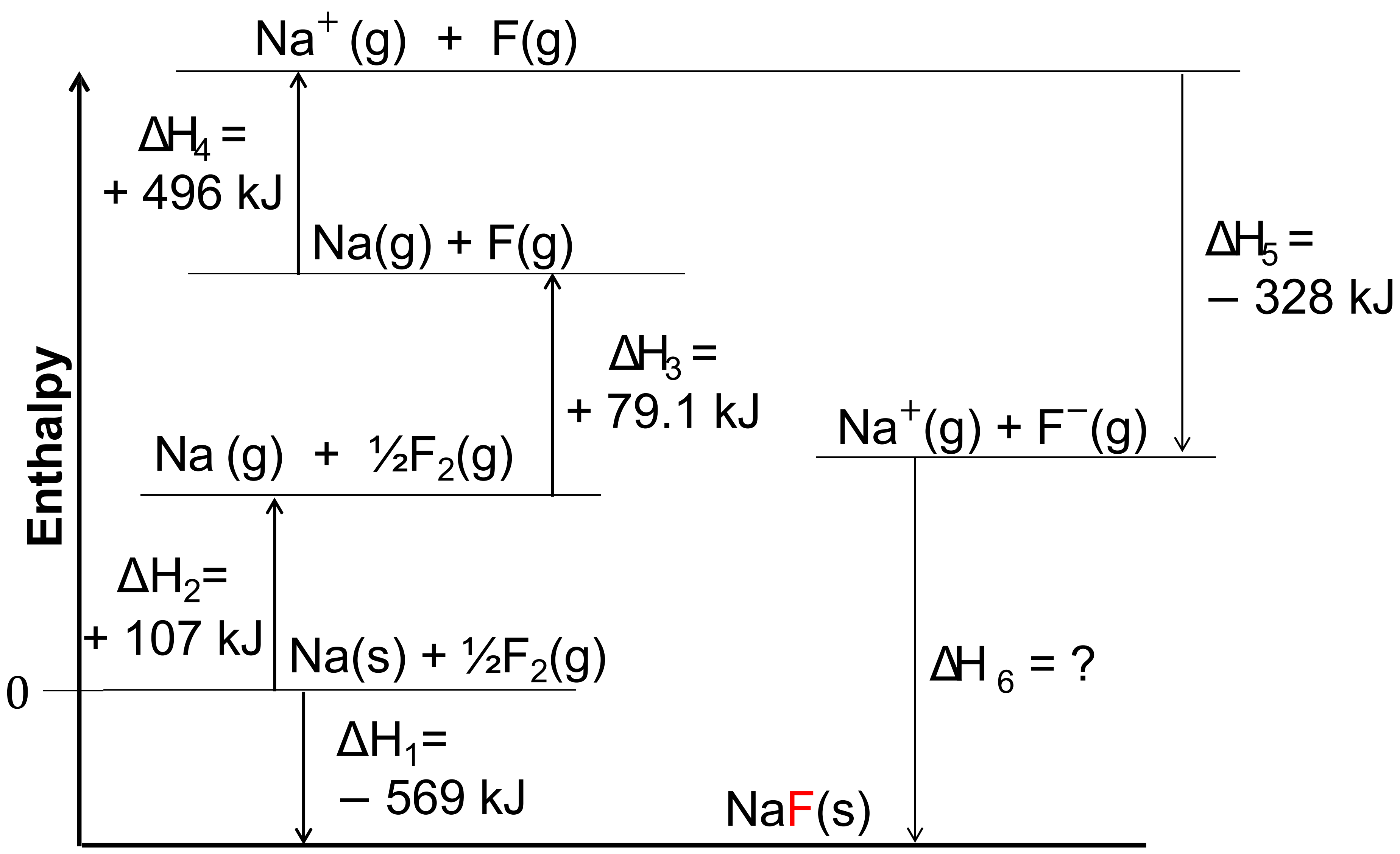
$$\Delta H_1 = \Delta H_2 + \Delta H_3 + \Delta H_4 + \Delta H_5 + \Delta H_6$$

$$-\ 569 = 107 + 496 + 79.1 + (-\ 328) + \Delta H_6$$

$$\mathbf{\Delta H_6 = -\ 923.1\ kJmol^{-1}}$$

Solution 2:

Using energy level diagram



$$\Delta H_1 = \Delta H_2 + \Delta H_3 + \Delta H_4 + \Delta H_5 + \Delta H_6$$

$$-569 = 107 + 496 + 79.1 + (-328) + \Delta H_6$$

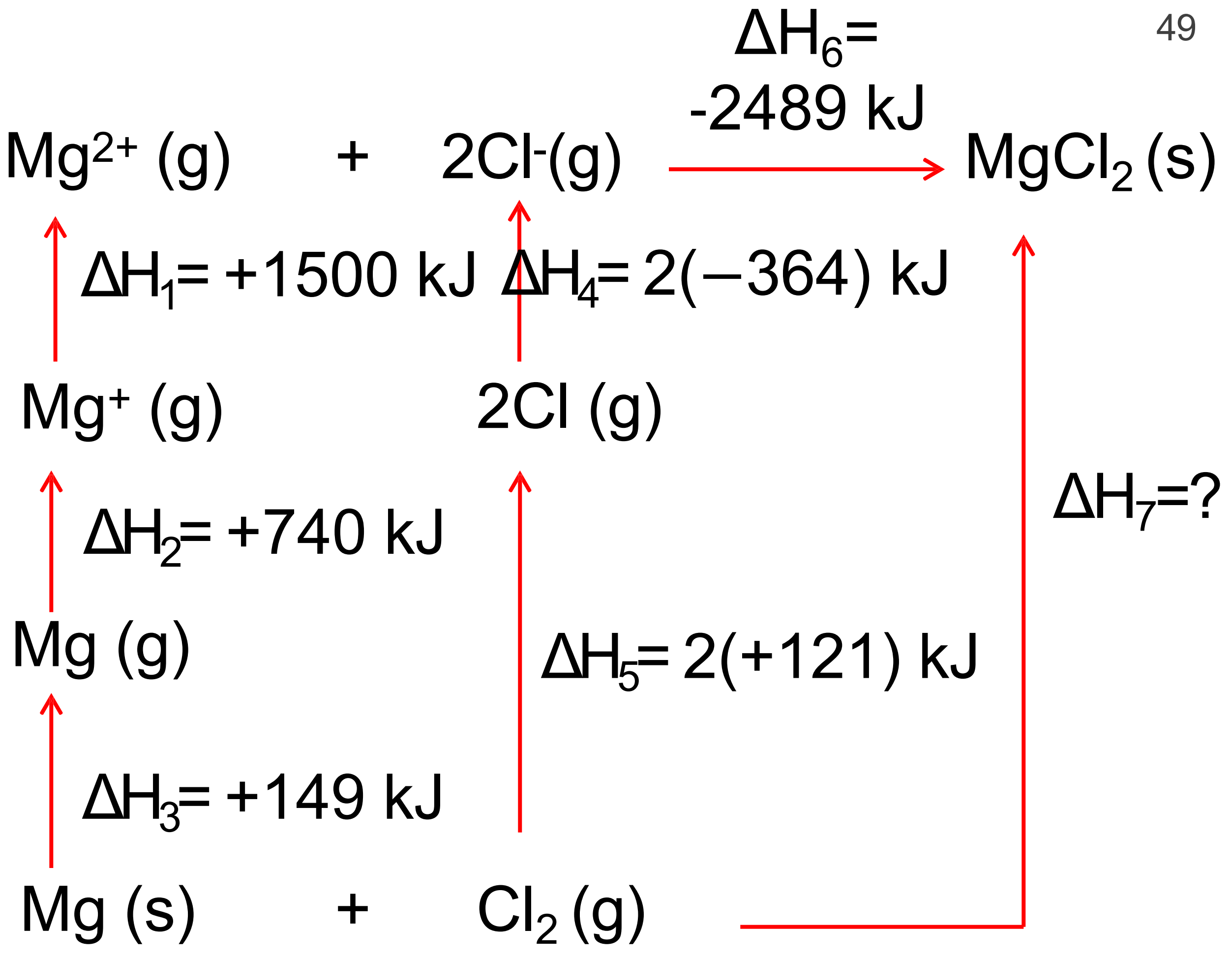
$$\mathbf{\Delta H_6 = -923.1\ kJmol^{-1}}$$

Example 2: Construct a Born-Haber cycle from the data below and determine the enthalpy of formation of **$MgCl_2$**.

	ΔH ($kJmol^{-1}$)
Standard enthalpy of atomisation of Mg	= +149
First ionisation energy of Mg	= +740
Second ionisation energy of Mg	= +1500
Standard enthalpy of atomisation of Cl	= +121
Electron affinity of Cl	= – 364
Lattice energy of $MgCl_2$	= – 2489

Solution:

Thermochemical equation	ΔH(kJmol^{-1})
$Mg(s) \rightarrow Mg(g)$	+149
$Mg(g) \rightarrow Mg^{+}(g) + e^{-}$	+740
$Mg^{+}(g) \rightarrow Mg^{2+}(g) + e^{-}$	+1500
$\frac{1}{2}Cl_2(g) \rightarrow Cl(g)$	+121
$Cl(g) + e^{-} \rightarrow Cl^{-}(g)$	– 364
$Mg^{2+}(g) + 2Cl^{-}(g) \rightarrow MgCl(s)$	– 2489
$Mg(s) + Cl_2(g) \rightarrow MgCl_2(s)$	ΔH_f



$$\Delta H_7 = \Delta H_1 + \Delta H_2 + \Delta H_3 + \Delta H_4 + \Delta H_5 + \Delta H_6$$

$$= 1500 + 740 + 149 + 2(-364) + 2(+121) + 2489$$

$$= -586$$

$$\mathbf{\Delta H_{formation} = -586\ kJmol^{-1}}$$

THE END